Project Overview

Project Title

Production Line Quality Risk Advisor

Problem Statement with Description

Manufacturing industries generate a large number of production observations involving different operating conditions such as temperature, rotational speed, torque, tool usage, and product characteristics. Quality-control teams have limited time and resources to manually inspect every production item, making it difficult to identify potentially defective or high-risk production conditions early.

The proposed system is a machine-learning-based quality risk assessment and decision-support application for manufacturing quality engineers. It will analyze production and operating conditions to identify observations that are more likely to be associated with machine failure or quality problems.

The system will provide an interactive interface where a quality engineer can examine the risk associated with production conditions, compare patterns with historical behavior, identify important factors contributing to risk, and prioritize observations for further inspection. The application will combine machine learning, statistical analysis, and interactive visualization rather than functioning only as a prediction model.

Objectives

- Develop a system to identify production observations with elevated machine-failure risk.
- Prioritize potentially risky observations when inspection capacity is limited.
- Analyze relationships between manufacturing operating conditions and failures.
- Apply appropriate machine-learning classification techniques and compare their performance with simple statistical baselines.
- Use statistical methods to validate relationships, distributions, correlations, and model performance.
- Provide understandable explanations of the factors contributing to predicted risk.
- Develop an interactive dashboard for quality engineers to explore production and risk patterns.
- Evaluate the system using suitable classification and decision-oriented performance metrics.

Scope

The project will focus on manufacturing quality-risk assessment and machine-failure classification using structured production data.

The system will include:

- Data preprocessing and handling of missing values and outliers.
- Exploratory Data Analysis and statistical analysis.

- Feature preparation and selection.
- Machine-failure risk classification.
- Model comparison and cross-validation.
- Risk prioritization based on limited inspection capacity.
- Model interpretation and feature-importance analysis.
- Interactive visualization of production conditions and failure patterns.
- A user-facing application for quality engineers.
- Testing, performance evaluation, documentation, and software-engineering artifacts.

The project will not attempt to control machinery, automatically shut down production equipment, or guarantee real-world machine safety. Its purpose is to provide decision support for quality-control and inspection activities.

Student Name

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